

SIH26249 — Master PRD (Whole Project)

SIH 2026 — SIH26249

Air Power: Predictive Maintenance & Fleet Availability

Ministry of Defence — Defence Services Staff College

SIH26249 — Air Power: Predictive Maintenance & Fleet Availability

00 — MASTER PRD (the whole project in one document)

Product name (working): READYFLEET — Fleet-Availability Decision Layer for Air Power **PS:** SIH26249 · "Air Power - Predictive Maintenance & Fleet Availability" · Ministry of Defence (MoD), Defence Services Staff College · Category: Software · Theme: Transportation & Logistics **Hackathon:** Smart India Hackathon 2026 · Build: 36-hour, laptop-local demo **Verified PS text** (opened 2026-10-03 at <https://sih2026-ps-viewer.vercel.app/ps/SIH26249>):

"Problem Statement: Low aircraft availability due to fragmented and largely reactive maintenance practices across the air fleet. Maintenance data from aircraft health-monitoring systems, technical records, spares and maintenance agencies is **not adequately integrated**, resulting in delayed fault prediction, avoidable aircraft downtime and sub-optimal utilisation of critical assets. Technology Opportunity: AI/ML-based predictive maintenance, IoT/aircraft health monitoring, digital twins and an integrated maintenance analytics platform."

USP (one line): The fleet-availability decision layer the PS says doesn't exist — fusing health predictions with spares, crew, and technical records to forecast mission-capable aircraft, and prescribing the maintenance action *today* that changes *tomorrow's* number.

Team profile: Vanilla JS + Canvas 2D, Python FastAPI microservices, Scikit-learn, Web Workers, MapLibre GL; digital-twin/simulation experience; demo runs on a laptop.

1. VISION & PS MAPPING

The PS names three distinct failure modes: (1) **fragmented, reactive maintenance**; (2) **data not adequately integrated** across health-monitoring systems, technical records, spares, and maintenance agencies; (3) outcomes: delayed fault prediction, avoidable downtime, sub-optimal asset utilisation. The Technology Opportunity names four building blocks: AI/ML predictive maintenance, IoT/aircraft health monitoring, digital twins, integrated maintenance analytics platform.

The trap most teams will fall into: build a better RUL predictor (an engine-health box). That solves failure mode (3) partially but ignores the PS's central diagnosis — **integration** — and the PS never asks for another sensor box. READYFLEET's answer:

PS phrase	Our mapping
"Low aircraft availability"	Fleet MC-rate forecast T+24h...T+7d is the product's headline number — the thing the commander looks at every morning.

PS phrase	Our mapping
"fragmented and largely reactive maintenance"	Reactive → predictive: RUL per component class feeds a forward-looking availability engine; fragmented → one fused decision layer.
"not adequately integrated" (health-monitoring, technical records, spares, agencies)	Five-stream fusion: (i) HUMS-style health signals, (ii) technical records, (iii) spares inventory + lead times, (iv) crew capacity, (v) sortie demand & mission priority. No single-stream tool does this.
"delayed fault prediction"	Per-component RUL with prediction intervals, alerted before the 24 h critical horizon.
"avoidable aircraft downtime"	NMC-driver ranking ("fix X → recovers Y tails") + what-if levers show exactly which actions avoid which downtime.
"sub-optimal utilisation of critical assets"	Mission-priority scoring; cannibalization advisor maximizes <i>mission-capable</i> hours, not raw airframe count.
Tech opportunity: AI/ML PdM, IoT/HUMS, digital twins, integrated analytics	AI/ML RUL (Scikit-learn GBM); health signals ingested as one input stream (we sit <i>above</i> HUMS, never compete with it); a scenario twin (the what-if engine replays cloned fleet states — a decision twin, not a physics twin); integrated analytics = the aggregator + dashboard.

Doctrinal anchors (USAF, used with mapping note): MC rate = (FMC hours + PMC hours) / possessed hours; NMC splits NMCM (maintenance) / NMCS (supply) / NMCB (both); CANN rate = cannibalization actions per 100 sorties, FY2000 USAF average **11.6** (per GAO-01-693T / GAO-02-86); the Langley/Elmendorf "Cannibalization Dock Program" centralizes donor management; urgency evidence: FY2024 B-52H **54%** MC, B-1B **43%** MC (North Star Readiness Campaign). NOTE: doctrine is US — the deck must say we *map* these to IAF maintenance terminology, not assume identical terms. Cannibalization framing per GAO: a *symptom of logistics shortfalls* — pitch as "authorized cannibalization with guardrails and a debt-repayment ledger," never as endorsing the practice. GAO report references: <https://www.gao.gov/assets/t-gao-01-693.pdf>

2. RESEARCH SUMMARY (10 papers — findings + gaps)

All sources in plain full URLs at the end (§13). Findings below; gaps feed directly into our differentiation.

1. **Li, Ding, Sun (2018), Reliability Engineering & System Safety** — first deep CNN predicting turbofan RUL directly from raw C-MAPSS sensor windows. *Gap: single-engine point prediction, no uncertainty, no fleet layer.*

2. **Özcan (2025), Scientific Reports** — LightGBM+CatBoost+GB ensemble on C-MAPSS, RMSE **6.62** on FD001, SHAP interpretability, $RUL \leq 15$ -cycle scheduler. *Gap: single-engine, no fleet aggregation, no spares/crew/mission.*
3. **Fan, Chang, Ji, Chen (2023), J. Advances in Information Fusion** — "learn-then-optimize": biLSTM multicomponent RUL + constrained MIP scheduling to maximize fleet availability; beats preventive maintenance *in simulation*. **Closest to our architecture.** *Gap: simulated only; omits spares inventory and sortie-demand forecasting; no uncertainty propagation.*
4. **Kozanidis, Liberopoulos, Pitsilkas (2010), Military Operations Research** — MILP flight-and-maintenance planning maximizing fleet availability, Hellenic Air Force case. *Gap: deterministic preventive logic, no prognostics.*
5. **Transformer-KAN-BiLSTM study (2025), Aerospace** — strong C-MAPSS accuracy; paper itself documents benchmark limits: synthetic data lacking sensor noise/real regimes, simple fault modes, no accessory gearbox/mounts/nacelle, overfitting blocks transfer to real fleets. (Authors not visible in render — do not cite named authors.) *Gap: point accuracy, no bridge to decisions.*
6. **Broer, Benedictus, Zarouchas (2022), Aerospace** — multi-sensor fusion framework for composite SHM (vibration, acoustic emission, guided waves, strain). *Gap: structures-only, conceptual.*
7. **Li & Li (2025), J. Applied Automation Technologies** — digital twin fusing physics-informed NN + Bayesian DL + virtual fault injection: **>97% fault detection, RUL error <8%, 21% maintenance cost reduction, 13% fleet availability improvement** on piston-engine GA aircraft (author-reported). *Gap: GA scale, synthetic, no military fleet/sortie/crew/spares.*
8. **Chen, Teng, Zhang, Gao (2026), Aerospace** — Markov queuing for rotatable spares: repair bottleneck doubles required stock 6→12 units at 95% service level; stock grows sublinearly with fleet size. *Gap: stationary failure stats, never ingests live PHM signals, isolated from MC rate.*
9. **Jardine, Lin, Banjevic (2006), Mechanical Systems & Signal Processing** — foundational CBM framework. *Gap: pre-ML, no fleet layer.*
10. **Saxena, Goebel, Simon, Eklund (2008), PHM08** — introduced the C-MAPSS turbofan degradation dataset (FD001–FD004, up to 260 training engines, 1–2 fault modes, 21 sensors). (Verified via secondary sources.) *Gap: the field's evidence base is this simulation — engine-only, no fleet context.*

GAPS WE OWN (the pitch's "nobody has built this"): - G1: Literature optimizes single-engine RUL accuracy while fleet availability is modeled only with deterministic preventive logic — nobody fuses live prognostics into a **forward mission-capable-rate forecast**. - G2: Fan et al. is closest but omits spares/crew/records/sortie demand. - G3: Spares research assumes stationary stats and no PHM ingestion. - G4: No published framework for **uncertainty-aware what-if maintenance levers** on readiness. - G5: Digital twins stay asset-level — the missing piece is the **fleet-availability decision layer**.

3. EXISTING SOLUTIONS × LIMITATIONS × OUR SOLUTIONS

Existing solution	Limitation (from public material)	Our solution
HAL HUMS (Dhruv ALH via IAC VMS; LUH "state-of-the-art HUMS") — vibration-based per-airframe diagnostics, post-flight download	Per-airframe component diagnostics; no fusion with spares/crew/records/demand; no fleet forecast	Sit ABOVE HUMS — consume its health signals as one input among five
GE Aerospace Collaborative Insight — 44K+ engines, 114M flight records/yr, 85%+ analytic effectiveness, 2000+ disruptions avoided (all <i>per GE's own page</i>)	Engine-centric, commercial, OEM ecosystem; no fleet MC forecast, no what-if levers	Operator-side, platform-agnostic fusion layer
Safran EngineLife — 60+ engine parameters → health indicators, daily indicators + monthly mission-tailored recommendations	OEM-locked to Safran engines	Cross-OEM fleet aggregation
DRDO digital-twin RFP (TDF 2025) — hybrid physics+AI engine twin for diagnostics	Asset-level engine twin; no delivered product evidenced; no fleet layer	Consume twin outputs as <i>inputs</i> ; solve the decision problem above the twin
IAF digital-twin push (Air Chief Marshal AP Singh, Kanpur, Aug 2026, <i>per press reports</i>) — predictive AI + digital twins	Statements of intent, warfare-scenario framing; no delivered maintenance tool	Build exactly the tool the strategy calls for — top-down demand validation
IFS Maintenix for Defense (US Navy N-MRO) — enterprise MRO system of record/workflow	Predictive is an add-on; no MC-rate forecast with levers; global suite not built for Indian operator	Decision-intelligence layer atop records systems; never requires ripping out the MRO system of record
Ramco Aviation MRO 6.0 (Chennai) — M&E/MRO suite + AI assistants (discrepancy clustering, SB extraction, demand forecasting)	Descriptive/assistive AI, not an availability decision layer; vendor metrics unverified	Add the missing optimization layer (MC forecast + levers)
Open-source C-MAPSS repos (e.g., karthik-1604 LSTM RMSE 15.42 / R ² 0.862 on FD001; ash180905/aerotwin-ai for SIH26054; turbofan-rul ~12.5 RMSE)	Single-engine RUL on simulated data; no fleet/fusion/levers	Treat RUL as the commodity baseline ; we decide fleets

Judge-facing honesty obligations on this table: competitor gaps come from *public product descriptions* — the deck must say "not advertised in their public product material," never "they lack it." Thin-evidence items: (a) no primary HAL source confirms an AI-RUL ground station — do not claim it; (b) Safran scale figures are ~2019 press-release era — never presented as current; (c) DRDO RFP scope details from defence-blog reporting only — existence medium, details thin; (d) AMCA "75% mission-capable rate" is a forum thread — **do not cite as an IAF target**; (e) Ramco AI metrics are vendor claims with no independent evaluation; (f) GE metrics are GE's own with no published methodology — use only with "per GE" attribution; (g) Air Chief statements from press reports, no primary transcript — say "per press reports"; (h) aerotwin-ai is a real SIH 2026 entry for a DIFFERENT PS (SIH26054) — related work, not a direct competitor; do not claim it does fleet availability.

4. FEATURE SPEC (P0/P1/P2 summary)

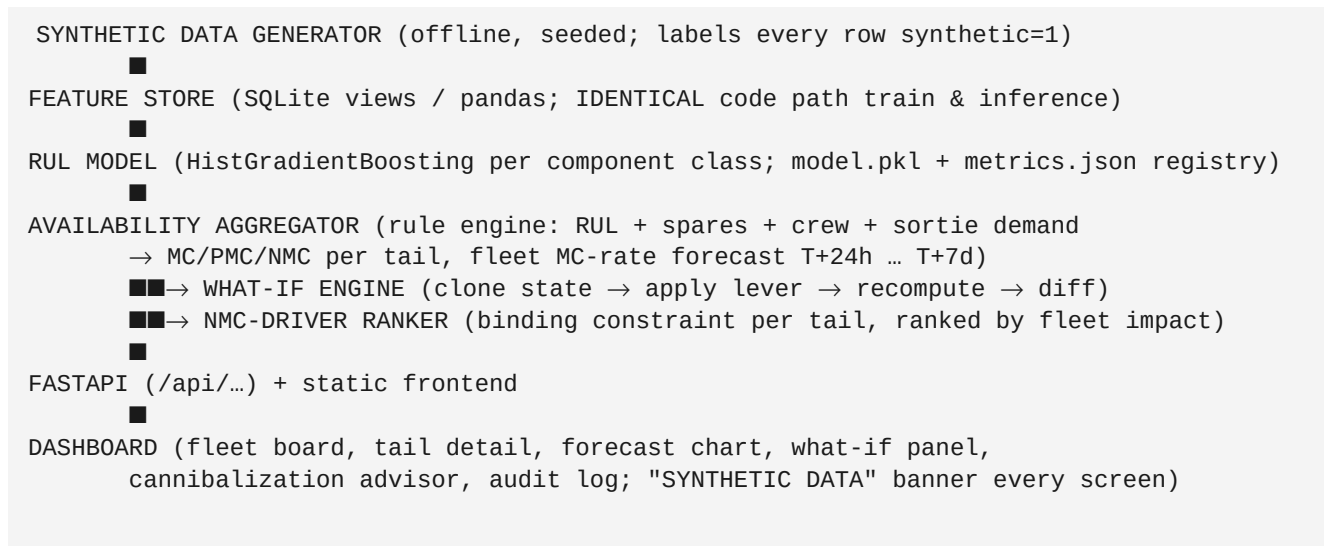
P0 (36h must-have): (1) Tomorrow's MC-Rate Forecaster [MEDIUM risk]; (2) What-If Maintenance Levers [MEDIUM]; (3) NMC-Driver Ranking [LOW]; (4) Cannibalization Trade-Off Advisor [MEDIUM — greedy net-MC heuristic, NOT full MILP]; (5) Sortie-Demand Fusion & Mission-Priority Scoring [LOW]; (6) Spares Shortfall / MICAP-Risk Predictor [MEDIUM]; (7) Per-Prediction Explainability Panel [LOW]; (8) Immutable Audit & Provenance Log [LOW]; (9) Crew-Capacity Constraint Scheduler [MEDIUM — scoped to 1-day greedy schedule]. P1 (stretch / demo-if-time): Per-Mission MC Forecast (FMC/PMC by mission profile) [LOW]; Cannibalization Debt Ledger [LOW]; Repeat/Recur Discrepancy Watchlist [LOW]; Break-Rate→Parts-Demand Predictor [MEDIUM]; Offline-First Base Deployment Story [MEDIUM — demo read-only offline]; Plain-English Daily Readiness Brief [LOW]; Deferred-Discrepancy Portfolio Optimizer [MEDIUM]. P2 (deck-only future work, explicitly NOT built): Back-Shop & Depot Queue Simulator [HIGH]; Federated Multi-Base Learning [HIGH]. Full acceptance criteria: 02-features-spec.md . Non-goals: competing sensor/HUMS hardware; full MILP/ILP optimizer; LSTM/Transformer RUL; real IAF data; multi-base federation; RBAC/auth; real-time streaming digital-twin physics.

5. TECH STACK (with rationale)

Layer	Choice	Rationale
Frontend	Vanilla JS + Canvas 2D SPA	Team's proven 60 Hz Canvas work; fleet board, cards, heatmaps, gauges all Canvas 2D — zero framework build cost in 36 h
Map	MapLibre GL, optional secondary tab only	Geographic view is secondary to the fleet board; keep it out of the critical path
Backend	Single Python FastAPI service (serves static frontend too; one uvicorn process)	One deployable, Pydantic validation, auto OpenAPI docs — backend + frontend in one process

Layer	Choice	Rationale
ML	Scikit-learn HistGradientBoostingRegressor for RUL	Trains in seconds on CPU; GBM-on-windowed-stats reaches RMSE in the same band as published LSTMs on FD001 (~13–20 vs ~12–18) while debuggable and live-retrainable. LSTM/Transformer explicitly NOT 36h-feasible
Simulation	Python batch NumPy/Pandas generator, seeded, pre-bakes full synthetic dataset before demo; tiny JS ticker replays sensor values for "live" feel	Deterministic; demo needs no live compute surprises
Database	SQLite single file (fleet ~40 tails is trivial)	Zero-ops; swap path to Postgres documented for future work
Deployment	Laptop-local uvicorn (Vercel backup only)	No cloud dependency during judging
Testing	pytest incl. model-eval script whose printout IS the judge evidence	Evidence, not just green tests

6. ARCHITECTURE (boxes → flow)



Full ASCII diagram, component responsibilities, API sketch: `03-architecture-techstack.md`.

7. DATA MODEL (summary)

SQLite tables: aircraft(tail_no PK, type, base, mission_priority 1-3, status cached) · component(id PK, tail_no FK, comp_class engine/avionics/hydraulics/airframe, serial_no, installed_cycles, installed_hours) · sensor_reading(id, component_id FK, cycle, op_setting_1..3, sensor_1..12, synthetic flag) · maintenance_record(id, tail_no, component_id, opened_at, closed_at, task_type scheduled/unscheduled, crew_hours, spare_used_id) · spare(part_no, comp_class, qty_on_hand, qty_due_in, lead_time_h, expedited_lead_time_h) · crew(id, name synthetic, trade, shift, available_h_per_day) · sortie(id, tail_no nullable, date, mission_type, priority, required_status) · prediction(id, component_id, run_id, predicted_rul_h, ci_low, ci_high, created_at) · scenario(id, name, base_run_id, lever_json, result_mc_rate, created_at) · audit_log(id, ts, actor, action, entity, entity_id, detail_json) . Full field specs, generator spec, ML pipeline, aggregator rules, what-if logic: 04-data-models-and-logic.md .

8. ALGORITHMS (plain words)

(a) Generator. Per-component failure mode; exponential degradation $\text{health}(t)=1-a \cdot \exp(b \cdot t)$ with randomized a, b per unit; sensors = baseline(health) + op-condition offsets (3 discrete settings, C-MAPSS-style) + Gaussian noise; RUL label = (failure_cycle – current_cycle) × hours_per_cycle; 5% infant-mortality units; censored no-fault units; maintenance events reset the curve; seed stored in metrics.json. **(b) RUL model.** Features = per-sensor rolling mean/std/min/max/slope over 10- and 30-cycle windows + cycle + cycles_since_overhaul + op-setting/comp-class one-hots; target RUL capped at 125 h (piecewise-linear trick); HistGradientBoostingRegressor with early stopping, validation split **BY TAIL** (no leakage); evaluate RMSE/MAE on held-out tails vs naive mean-RUL baseline; prediction intervals via quantile GBM or $\pm 1.5 \times$ validation MAE. **(c) Aggregator.** Tail NMC on day d if ANY of: critical component RUL < 24 h; component RUL < spare.lead_time_h AND qty_on_hand == 0; required fix crew_hours > available trade crew-hours in window. States MC/PMC/NMC. Fleet MC rate(d) = MC tails / total. Unfilled high-priority sorties = headline capability-gap number. All thresholds in one config file. **(d) What-if.** Lever = pure state patch on cloned snapshot (expedite spares → lead_time := expedited; reassign crew → move N hours between trades; cannibalize → donor NMC-down-for-parts, recipient gets donor part RUL; donor must be lower mission priority AND already NMC); re-run aggregator (milliseconds); return {mc_rate_before, mc_rate_after, tails_changed, unfilled_sorties before/after}; store each run as a scenario row. **(e) NMC-driver ranking.** Binding constraint per NMC tail; fleet score = (tails grounded by driver) × (avg mission priority) × (days grounded in 7-day forecast); top-5 presented as "fix X → recovers Y tails."

9. HONESTY & DATA RULES (NON-NEGOTIABLE — repeated in the demo doc)

- 1. ALL data is synthetic.** "SYNTHETIC DATA" banner on **every screen**; every API response includes "data_class": "synthetic" .
- 2. Never imply real IAF fleet data.** No roundels, no real tail numbers, no squadron names that map to real units.
- 3. Never claim "worldwide first"** — the classified defence space is unknowable; claim only what the public evidence supports ("not advertised in their public product material").

4. **We are the decision layer, never a competing sensor/HUMS box.** HAL's HUMS does per-platform diagnostics; we sit ABOVE it and consume its signals.
5. **Respect the thin-evidence list (§3) in every judge-facing claim.** GE numbers always "per GE"; Air Chief quotes always "per press reports"; never cite the AMCA-forum 75% figure.
6. **Measured vs estimate discipline:** report only measured/sourced metrics; mark every estimate as **ESTIMATE**.
7. Automated enforcement: tests assert every DB row has synthetic=1 and every API response carries data_class=synthetic; UI banner presence test.

10. 36-HOUR BUILD PLAN (summary)

(1) generator + schema + seed determinism → (2) features + GBM train/eval (get RMSE-vs-baseline printout EARLY) → (3) aggregator + what-if + driver ranker as pure Python + unit tests → (4) FastAPI endpoints → (5) Canvas dashboard (fleet board → tail drill-down → what-if panel, in that order) → (6) scripted demo scenario + honesty banners. Numbered tasks with acceptance criteria:

`05-build-plan-tasks.md` .

11. DEMO SCRIPT (summary) & JUDGE Q&A ARMOURY (summary)

5-minute demo, minute-by-minute, plus the 10 toughest questions with one-line rebuttals and on-screen honesty badges: `06-demo-and-judge-qa.md` . Closing line: **"We don't predict engine failure — we tell the commander how many aircraft fly tomorrow."**

12. RISKS

- **Fusion-calibration risk (P0 feature 1, MEDIUM):** fusing 5 streams into one calibrated MC number is the hardest modeling task → mitigation: ship the rule-based aggregator first (deterministic, demoable), treat any learned calibration as stretch.
- **Cannibalization heuristic defensibility:** greedy net-MC, not MILP → stated openly; guardrails (priority + CANN-rate limits + debt ledger) carry the story.
- **Synthetic-data credibility:** judges may discount results → counter with determinism (same seed → identical SHA-256), the documented generator, and the explicit integration roadmap to real feeds.
- **Scope creep into physics twins:** any physics-level twin work is out; the what-if "scenario twin" is the only twin we build.
- **Laptop-demo fragility:** pre-bake everything; the JS ticker is replay, stated openly; Vercel as backup.

13. FUTURE WORK

P2 deck-only: Back-Shop & Depot Queue Simulator; Federated Multi-Base Learning. Plus: SQLite→Postgres swap path; OR-Tools MILP for the cannibalization advisor (stretch); learned availability calibration; integration roadmap into IAF maintenance IT (Maintenix/Ramco/HUMS feeds); survival-analysis RUL as a modeling upgrade.

14. SOURCES (plain full URLs — copy-paste safe)

PS page: <https://sih2026-ps-viewer.vercel.app/ps/SIH26249>

Papers:

https://escholarship.org/content/qt5ns8r3fs/qt5ns8r3fs_noSplash_7d463f0d432194b63917f89e056f2644.pdf?t=pihyfp

<https://www.nature.com/articles/s41598-025-23473-2>

<https://isif.org/files/isif/2024-04/06-102022-0013R1.pdf> <https://www.jstor.org/stable/43941217>

<https://www.mdpi.com/2226-4310/12/11/998> <https://www.mdpi.com/2226-4310/9/4/183>

<http://www.wpias.edu.pl/ojs/index.php/jaat/article/view/69>

https://mdpi-res.com/d_attachment/aerospace/aerospace-13-00620/article_deploy/aerospace-13-00620-v2.pdf?version=1

<https://bibbase.org/network/publication/jardine-lin-banjevic-areviewonmachinerydiagnosticsandprognosticsimplementi>

<https://github.com/senanurcetin/smart-factory-app/blob/HEAD/docs/rul-case-study.md>

Doctrine & context: <https://www.gao.gov/assets/t-gao-01-693.pdf>